Roberto Nájera

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Computer Science student focused on backend, full-stack and AI-powered systems (LLM agents, MCP, PyTorch/LoRA). First place in Guatemala at the ICPC Gran Premio Centroamericano 2026.

Education

Universidad del Valle de Guatemala — Guatemala City, Guatemala

Bachelor of Science in Computer Science and Information Technology — *Jan. 2023 – Expected Dec. 2027*

- Awarded Highest Academic Distinction each academic year since enrollment.
- Strong foundation in algorithms, data structures, probability theory, and discrete mathematics.

Awards and Competitions

International Collegiate Programming Competition (ICPC) — 2025 – 2026

Latin American Regional — Gran Premio Centroamericano — *ICPC Profile*

- First place in Guatemala (out of 60 teams; 124 across Central America) at the 2026 Gran Premio Centroamericano, held within the ICPC Latin American Regional, and top-ranked team from Universidad del Valle de Guatemala.
- Second place in Guatemala (out of 36 teams; 93 across Central America) at the 2025 Gran Premio Centroamericano and top-ranked team from Universidad del Valle de Guatemala.

National Champion — Copa Centroamericana de Programación — 2023

Hosted by OmegaUP

Silver medal in Olimpiada Iberoamericana de Biología — 2022

Best result ever achieved by Guatemala to date.

Silver medal in Biology — Olimpiada Nacional de Ciencias USAC — 2022

Also qualified to advanced rounds in the Math Olympiads during 2021.

Experience

Full Stack Junior Dev — Feb. 2026 – June 2026

Pacifiko.com — *PHP, OpenCart, MySQL, Python, AWS, Jenkins*

- Integrated AI agent pipelines in Python to automate catalog enrichment (several thousand products, after filtering out low-quality ones), reducing hundreds of hours of manual work and improving product attribute coverage.
- Designed and implemented the Extended Warranty feature: defined its full architecture and refactored legacy warranty logic into a scalable solution, built for electronics such as computers and phones and configurable for any catalog category.
- Designed the AWS architecture for a planned BNPL (buy now, pay later) fintech platform, integrating external providers for the core lender (accounts and contracts), identity verification and eligibility analytics, along with the databases, the checkout widget on the storefront, NAT configuration and access permissions.

Undergraduate Teaching Assistant — Jan. 2025 – June 2026

Universidad del Valle de Guatemala

- Served as a Teaching Assistant for the courses Introduction to IT, Programming Fundamentals, and Probability Theory.
- Assisted in grading, project evaluation, and conceptual reinforcement of core CS and math topics for 100+ students across the different courses.
- Provided special mentoring and Calculus tutoring to struggling students, who later successfully completed the course.

Spring Bootcamp Ada School — March 2024 – June 2024

Proyecto Innova - Swisscontact

- Completed a Spring bootcamp focused on backend design, RESTful APIs, authentication and database systems.
- Developed practice projects following ACID principles and layered architecture to reinforce good software engineering practices.
- Implemented services such as JWT authentication and both SQL and NoSQL database connections, enabling secure authentication and reliable database interactions.

Mentorship program "Juntos Llenamos la Copa" May 2023 – Oct. 2023
OmegaUP

- Selected for a mentorship program as a prize for top performance in a programming competition.
- Mentored by engineers from Google, focused on software skills and Big Tech work environments.
- Gained exposure to industry expectations and received technical advice to strengthen my skills and professional performance.

Projects

DesignBetter GitHub Link | *Django, Next.js, PostgreSQL, Docker, Vercel* February 2025 – December 2025

- Delivered a full-stack e-commerce web application for fashion designs, using Django to serve a REST API and Next.js with React for the frontend, tailored to the workflows of fashion designers.
- Served as Scrum Team Manager, delegating tasks and story points across a team of four developers; completed 100+ story points while practicing Lean Software Development.
- Implemented secure authentication flows with Google OAuth and JWT-based sessions to handle sign-up and login reliably.
- Integrated Three.js for interactive 3D garment visualization, improving user engagement with product previews.
- Built and deployed a CI/CD pipeline using Vercel, enabling automatic builds and deployments on every change merged into the main Git branch.

QuantDesk + quant-mcp GitHub Link | *Python, MCP, Anthropic API, pandas, SciPy* Sept. 2026

- Developed QuantDesk, a console chatbot for quantitative portfolio analysis that uses Claude (Anthropic API) as the coordinating LLM and the Model Context Protocol (MCP) to call tools for financial calculations, file access and Git version control.
- Implemented quant-mcp, a custom stdio MCP server with tools for volatility, Sharpe ratio, historical and parametric VaR, correlation matrices and rebalance simulation, computed with pandas, NumPy and SciPy so that no result is estimated by the LLM.
- Built the host with an MCP client manager that discovers the tools of multiple servers, routes each call to the right server and logs all JSON-RPC traffic.

LoRA vs. Fine-Tuning vs. Prompting GitHub Link | *PyTorch, Hugging Face* Oct. 2026 – Nov. 2026

- Compared three language model adaptation strategies (zero-shot and few-shot prompting with Qwen2.5-1.5B, full fine-tuning, and LoRA on XLM-RoBERTa base) for sentiment analysis on Spanish tweets, evaluated on 870 test tweets using a free T4 GPU.
- LoRA (r=8) matched full fine-tuning in macro F1 (0.673 vs. 0.667) while training 0.89M of 278M parameters (about 0.3%), with 1.9 GB vs. 5.4 GB peak GPU memory and 33 s vs. 422 s of training time.
- Evaluated with 5 seeds (mean ± standard deviation), ablations on LoRA rank and target modules, and SHAP error analysis, avoiding data leakage by excluding a model pretrained on the same tweets.

Technical Skills

Languages: Python, C/C++, SQL (PostgreSQL, MySQL), Java, JavaScript, TypeScript, Rust, PHP, Kotlin, Go, R, HTML/CSS
AI and Data: PyTorch, Hugging Face, LoRA, LLM agents, Model Context Protocol (MCP), pandas, NumPy, SciPy, Matplotlib, SageMath
Frameworks: Django, Next.js, React, Node.js, Spring Boot, OpenCart, Vite, JUnit, Material-UI
Cloud and DevOps: AWS, Google Cloud Platform, Docker, Jenkins, CI/CD (Vercel), Git, Cursor
Parallel Computing, Compilers and Graphics: OpenMP, MPI, CUDA, ANTLR, Three.js, Raylib
Languages Spoken: Spanish (native), English (C1)